ii. Hold the yarn horizontally with the pair of tweezers or tongs. Feed the fibres slowly into the edge of the flame and observe what happens. Repeat this several times to check results.



ACTIVITY 5

Collect few natural and man made fibres and observe the burning characteristic by flame test.

Burning characteristics of Fibres.

Fibres	Cotton	Silk	Wool	Nylon	Polyester
Approaching flame	Ignites upon contact, does not shrink away	Curls away from flame	Curls away from flame	Melts away from flame, Shrink and fuses.	Fuses, melts and shrink away from flame
In flame	Burns quickly	Burn slowly and sputters	Burns slowly	Burns slowly with melting	Burns slowly and continuous to melt
Removed from flame	Continues burning, after glow	Usually self-extinguish	self-extinguish	self-extinguish	self-extinguish
Odor	Similar to burning paper	Similar to burning hair	Similar to burning hair	Celery	Chemical odor
Residue	Light, feathery ash grey in colour	Crushable bead, black in colour	Brittle, small black bead	Hard, tough, grey tan bead	Hard, tough, black or brown bead

4.4 Yarn Making

Yarns play an important role in the fabric manufacturing process . Yarn is "a generic term for a continuous strand of textile fibers, filaments, or material in a form suitable for knitting, weaving, or otherwise intertwining to form a textile fabric. Yarns are also used for products such as sewing and embroidery thread, string, and rope. Yarns are produced in various sizes and textures, and also vary in other characteristics. Performance, end use, and fabric care are affected by these yarn characteristics.

4.4.1 Yarn Twist

Yarns are twisted to hold the fibres together. The number of twists per unit length is used to measure twist. Yarn twist can be broadly divided by number of twists: none

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